

Day4Practice Sheet — PandasDataFrames&

Module: NumPy& Pandas — PandasDataFrames &Filtering

PreparedBy: Mr.Akarsh/Dr.RavneetKaur

Objective: PractisecreatingDataFrames,loadingCSVs,usinginfo()/describe()),loc,iloc,andbooleanfiltering.

SAMPLEDATASET

Use the Students dataset below (copy into a CSV file named 'students.csv') OR create it directly as a DataFrame using `pd.DataFrame(dict)`.

StudentID	Name	Age	City	Department	Score	Grade	Passed
S001	Arjun Mehta	21	Delhi	CSE	78	B	TRUE
S002	Divya Sharma	22	Mumbai	ECE	91	A	TRUE
S003	Karan Patel	20	Pune	CSE	55	C	TRUE
S004	Meera Nair	23	Chennai	MECH	42	D	TRUE
S005	Sahil Gupta	21	Delhi	ECE	88	A	TRUE

S006	Anjali Singh	22	Jaipur	CSE	35	F	FALSE
S007	Rohan Verma	20	Mumbai	MECH	67	B	TRUE
S008	Pooja Yadav	23	Chennai	CSE	92	A	TRUE
S009	Vivek Kumar	21	Pune	ECE	71	B	TRUE
S010	Nisha Agarwal	22	Delhi	MECH	58	C	TRUE

PRACTICE QUESTIONS

Question	Task Description	YourAnswer /Code																																																																																																												
Q1[Load&Inspect]	CreatetheDataFramefrom he table above. Display:shape, columns, dtypes,head(5), and tail(3).	Q1 [Load & Inspect] Task: Display shape, columns, dtypes, head(5), and tail(3). <pre>print("Shape:", df.shape) print("\nColumns:", df.columns) print("\nData Types:\n", df.dtypes) print("\nHead(5):\n", df.head(5)) print("\nTail(3):\n", df.tail(3))</pre> Shape: (10, 8) Columns: Index(['StudentID', 'Name', 'Age', 'City', 'Department', 'Score', 'Grade', 'Passed'], dtype='object') Data Types: <table><tr><td>StudentID</td><td>int64</td></tr><tr><td>Name</td><td>object</td></tr><tr><td>Age</td><td>int64</td></tr><tr><td>City</td><td>object</td></tr><tr><td>Department</td><td>object</td></tr><tr><td>Score</td><td>int64</td></tr><tr><td>Grade</td><td>object</td></tr><tr><td>Passed</td><td>bool</td></tr><tr><td>dtype:</td><td>object</td></tr></table> Head(5): <table><tr><th></th><th>StudentID</th><th>Name</th><th>Age</th><th>City</th><th>Department</th><th>Score</th><th>Grade</th><th>Passed</th></tr><tr><td>0</td><td>5001</td><td>Arjun Mehta</td><td>21</td><td>Delhi</td><td>CSE</td><td>78</td><td>B</td><td>True</td></tr><tr><td>1</td><td>5002</td><td>Divya Sharma</td><td>22</td><td>Mumbai</td><td>ECE</td><td>91</td><td>A</td><td>True</td></tr><tr><td>2</td><td>5003</td><td>Karan Patel</td><td>20</td><td>Pune</td><td>CSE</td><td>55</td><td>C</td><td>True</td></tr><tr><td>3</td><td>5004</td><td>Meera Nair</td><td>23</td><td>Chennai</td><td>MECH</td><td>42</td><td>D</td><td>True</td></tr><tr><td>4</td><td>5005</td><td>Sahil Gupta</td><td>21</td><td>Delhi</td><td>ECE</td><td>88</td><td>A</td><td>True</td></tr></table> Tail(3): <table><tr><th></th><th>StudentID</th><th>Name</th><th>Age</th><th>City</th><th>Department</th><th>Score</th><th>Grade</th><th>Passed</th></tr><tr><td>7</td><td>5008</td><td>Pooja Yadav</td><td>23</td><td>Chennai</td><td>CSE</td><td>92</td><td>A</td><td>True</td></tr><tr><td>8</td><td>5009</td><td>Vivek Kumar</td><td>21</td><td>Pune</td><td>ECE</td><td>71</td><td>B</td><td>True</td></tr><tr><td>9</td><td>5010</td><td>Nisha Agarwal</td><td>22</td><td>Delhi</td><td>MECH</td><td>58</td><td>C</td><td>True</td></tr></table>	StudentID	int64	Name	object	Age	int64	City	object	Department	object	Score	int64	Grade	object	Passed	bool	dtype:	object		StudentID	Name	Age	City	Department	Score	Grade	Passed	0	5001	Arjun Mehta	21	Delhi	CSE	78	B	True	1	5002	Divya Sharma	22	Mumbai	ECE	91	A	True	2	5003	Karan Patel	20	Pune	CSE	55	C	True	3	5004	Meera Nair	23	Chennai	MECH	42	D	True	4	5005	Sahil Gupta	21	Delhi	ECE	88	A	True		StudentID	Name	Age	City	Department	Score	Grade	Passed	7	5008	Pooja Yadav	23	Chennai	CSE	92	A	True	8	5009	Vivek Kumar	21	Pune	ECE	71	B	True	9	5010	Nisha Agarwal	22	Delhi	MECH	58	C	True
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Q2 [describe/info]	Run df.info() and df.describe(). From describe(): what is the mean Score? What is the standard deviation?	Q2 [describe/info] Task: Run df.info() and df.describe(). What is the mean Score? What is the standard deviation? <pre>df.info() print(df.describe()) # Answering the specific questions: mean_score = df["Score"].mean() std_score = df["Score"].std() print(f"Mean Score: {mean_score}") print(f"Standard Deviation of Score: {std_score}")</pre><pre><class 'pandas.core.frame.DataFrame'> RangeIndex: 10 entries, 0 to 9 Data columns (total 8 columns): # Column Non-Null Count Dtype --- - - 0 StudentID 10 non-null int64 1 Name 10 non-null object 2 Age 10 non-null int64 3 City 10 non-null object 4 Department 10 non-null object 5 Score 10 non-null int64 6 Grade 10 non-null object 7 Passed 10 non-null bool dtypes: bool(1), int64(3), object(4) memory usage: 792.0+ bytes</pre><table><thead><tr><th></th><th>StudentID</th><th>Age</th><th>Score</th></tr></thead><tbody><tr><td>count</td><td>10.00000</td><td>10.00000</td><td>10.00000</td></tr><tr><td>mean</td><td>5005.50000</td><td>21.50000</td><td>67.70000</td></tr><tr><td>std</td><td>3.02705</td><td>1.00012</td><td>20.13316</td></tr><tr><td>min</td><td>5001.00000</td><td>20.00000</td><td>35.00000</td></tr><tr><td>25%</td><td>5003.25000</td><td>21.00000</td><td>55.75000</td></tr><tr><td>50%</td><td>5005.50000</td><td>21.50000</td><td>69.00000</td></tr><tr><td>75%</td><td>5007.75000</td><td>22.00000</td><td>85.50000</td></tr><tr><td>max</td><td>5010.00000</td><td>23.00000</td><td>92.00000</td></tr></tbody></table> Mean Score: 67.7 Standard Deviation of Score: 20.13316776973868		StudentID	Age	Score	count	10.00000	10.00000	10.00000	mean	5005.50000	21.50000	67.70000	std	3.02705	1.00012	20.13316	min	5001.00000	20.00000	35.00000	25%	5003.25000	21.00000	55.75000	50%	5005.50000	21.50000	69.00000	75%	5007.75000	22.00000	85.50000	max	5010.00000	23.00000	92.00000
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Q3[Column Access]	Access the 'Name' column using df['Name']. Then access 'Score' and 'Grade' columns together using double brackets.	Q3 [Column Access] Task: Access 'Name' using single brackets, then 'Score' and 'Grade' using double brackets. <pre>name_col = df["Name"] score_grade_cols = df[["Score", "Grade"]]</pre>																																				
Q4 [loc]	Use df.loc to select: (a) row at index 0, (b) rows 2 to 5, (c) rows 0–4 but only columns Name, Score, Grade.	Q4 [loc] Task: Use df.loc to select specific rows and columns by label. <pre>row_0 = df.loc[0] rows_2_to_5 = df.loc[2:5] rows_cols_loc = df.loc[0:4, ["Name", "Score", "Grade"]]</pre>																																				
Q5 [iloc]	Use df.iloc to select: (a) first 3 rows, (b) last 2 rows, (c) rows at positions 1, 3, 5 and columns at positions 0, 2, 4.	Q5 [iloc] Task: Use df.iloc to select by integer position <pre>first_3 = df.iloc[0:3] last_2 = df.iloc[-2:] specific_iloc = df.iloc[[1, 3, 5], [0, 2, 4]]</pre>																																				

Q6[BooleanFilter]	Filter students where Score > 80. How many students scored above 80?	Q6 [Boolean Filter] Task: Filter students where Score > 80. How many students scored above 80? <pre>above_80 = df[df["Score"] > 80] print(above_80) print(f"Number of students who scored above 80: {len(above_80)}")</pre><table><thead><tr><th>StudentID</th><th>Name</th><th>Age</th><th>City</th><th>Department</th><th>Score</th><th>Grade</th><th>Passed</th></tr></thead><tbody><tr><td>1</td><td>5002</td><td>Divya Sharma</td><td>22</td><td>Mumbai</td><td>ECE</td><td>91</td><td>A</td><td>True</td></tr><tr><td>4</td><td>5005</td><td>Sahil Gupta</td><td>21</td><td>Delhi</td><td>ECE</td><td>88</td><td>A</td><td>True</td></tr><tr><td>7</td><td>5008</td><td>Pooja Yadav</td><td>23</td><td>Chennai</td><td>CSE</td><td>92</td><td>A</td><td>True</td></tr></tbody></table> Number of students who scored above 80: 3 ** Answer: 3 students (Divya, Sahil, and Pooja).**	StudentID	Name	Age	City	Department	Score	Grade	Passed	1	5002	Divya Sharma	22	Mumbai	ECE	91	A	True	4	5005	Sahil Gupta	21	Delhi	ECE	88	A	True	7	5008	Pooja Yadav	23	Chennai	CSE	92	A	True																																																															
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Q7[BooleanFilter]	Filter students from Delhi who are in CSE department. Show Name, Score, and Grade only.	Q7 [Boolean Filter] Task: Filter students from Delhi in CSE. Show Name, Score, and Grade only. <pre>delhi_cse = df[(df["City"] == "Delhi") & (df["Department"] == "CSE")] result_q7 = delhi_cse[["Name", "Score", "Grade"]] print(result_q7)</pre><table><thead><tr><th>Name</th><th>Score</th><th>Grade</th></tr></thead><tbody><tr><td>0</td><td>Anjun Mehta</td><td>78</td><td>B</td></tr></tbody></table>	Name	Score	Grade	0	Anjun Mehta	78	B																																																																																											
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Q8[BooleanFilter]	Find all students who Failed (Passed == False). What is their average Score?	Q8 [Boolean Filter] Task: Find all students who Failed (Passed == False). What is their average Score? <pre>failed_students = df[df["Passed"] == False] avg_failed_score = failed_students["Score"].mean() print(failed_students) print(f"Average Score of failed students: {avg_failed_score}")</pre><table><thead><tr><th>StudentID</th><th>Name</th><th>Age</th><th>City</th><th>Department</th><th>Score</th><th>Grade</th><th>Passed</th></tr></thead><tbody><tr><td>5</td><td>5006</td><td>Anjali Singh</td><td>22</td><td>Jaipur</td><td>CSE</td><td>35</td><td>F</td><td>False</td></tr></tbody></table> Average Score of failed students: 35.0 ** Answer: Only 1 student failed (Anjali), so the average score is 35.0.**	StudentID	Name	Age	City	Department	Score	Grade	Passed	5	5006	Anjali Singh	22	Jaipur	CSE	35	F	False																																																																																	
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Q9[String Filter]	Filter students whose Name contains the string 'a' (case-insensitive) using .str.contains().	Q9 [String Filter] Task: Filter students whose Name contains the string 'a' (case-insensitive) <pre>contains_a = df[df["Name"].str.contains("a", case=False)] print(contains_a)</pre><table><thead><tr><th>StudentID</th><th>Name</th><th>Age</th><th>City</th><th>Department</th><th>Score</th><th>Grade</th><th>Passed</th></tr></thead><tbody><tr><td>0</td><td>5001</td><td>Anjun Mehta</td><td>21</td><td>Delhi</td><td>CSE</td><td>78</td><td>B</td><td>True</td></tr><tr><td>1</td><td>5002</td><td>Divya Sharma</td><td>22</td><td>Mumbai</td><td>ECE</td><td>91</td><td>A</td><td>True</td></tr><tr><td>2</td><td>5003</td><td>Karan Patel</td><td>20</td><td>Pune</td><td>CSE</td><td>55</td><td>C</td><td>True</td></tr><tr><td>3</td><td>5004</td><td>Meera Nair</td><td>23</td><td>Chennai</td><td>MECH</td><td>42</td><td>D</td><td>True</td></tr><tr><td>4</td><td>5005</td><td>Sahil Gupta</td><td>21</td><td>Delhi</td><td>ECE</td><td>88</td><td>A</td><td>True</td></tr><tr><td>5</td><td>5006</td><td>Anjali Singh</td><td>22</td><td>Jaipur</td><td>CSE</td><td>35</td><td>F</td><td>False</td></tr><tr><td>6</td><td>5007</td><td>Rohan Verma</td><td>20</td><td>Mumbai</td><td>MECH</td><td>67</td><td>B</td><td>True</td></tr><tr><td>7</td><td>5008</td><td>Pooja Yadav</td><td>23</td><td>Chennai</td><td>CSE</td><td>92</td><td>A</td><td>True</td></tr><tr><td>8</td><td>5009</td><td>Vivek Kumar</td><td>21</td><td>Pune</td><td>ECE</td><td>71</td><td>B</td><td>True</td></tr><tr><td>9</td><td>5010</td><td>Nisha Agarwal</td><td>22</td><td>Delhi</td><td>MECH</td><td>58</td><td>C</td><td>True</td></tr></tbody></table>	StudentID	Name	Age	City	Department	Score	Grade	Passed	0	5001	Anjun Mehta	21	Delhi	CSE	78	B	True	1	5002	Divya Sharma	22	Mumbai	ECE	91	A	True	2	5003	Karan Patel	20	Pune	CSE	55	C	True	3	5004	Meera Nair	23	Chennai	MECH	42	D	True	4	5005	Sahil Gupta	21	Delhi	ECE	88	A	True	5	5006	Anjali Singh	22	Jaipur	CSE	35	F	False	6	5007	Rohan Verma	20	Mumbai	MECH	67	B	True	7	5008	Pooja Yadav	23	Chennai	CSE	92	A	True	8	5009	Vivek Kumar	21	Pune	ECE	71	B	True	9	5010	Nisha Agarwal	22	Delhi	MECH	58	C	True
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Q10[Add Column]	Add a new column 'Score_Category': if score >= 80 → 'Distinction', Score >= 60 → 'Merit', else 'Pass/Fail'. Use np.where() or nested condition.	Q10 [Add Column] Task: Add a new column 'Score_Category' using np.where() or nested conditions. <pre>conditions = [df["Score"] >= 80, df["Score"] >= 60] choices = ["Distinction", "Merit"] df["Score_Category"] = np.select(conditions, choices, default="Pass/Fail") print(df[["Name", "Score", "Score_Category"]])</pre><table><thead><tr><th>Name</th><th>Score</th><th>Score_Category</th></tr></thead><tbody><tr><td>0</td><td>Anjun Mehta</td><td>78</td><td>Merit</td></tr><tr><td>1</td><td>Divya Sharma</td><td>91</td><td>Distinction</td></tr><tr><td>2</td><td>Karan Patel</td><td>55</td><td>Pass/Fail</td></tr><tr><td>3</td><td>Meera Nair</td><td>42</td><td>Pass/Fail</td></tr><tr><td>4</td><td>Sahil Gupta</td><td>88</td><td>Distinction</td></tr><tr><td>5</td><td>Anjali Singh</td><td>35</td><td>Pass/Fail</td></tr><tr><td>6</td><td>Rohan Verma</td><td>67</td><td>Merit</td></tr><tr><td>7</td><td>Pooja Yadav</td><td>92</td><td>Distinction</td></tr><tr><td>8</td><td>Vivek Kumar</td><td>71</td><td>Merit</td></tr><tr><td>9</td><td>Nisha Agarwal</td><td>58</td><td>Pass/Fail</td></tr></tbody></table>	Name	Score	Score_Category	0	Anjun Mehta	78	Merit	1	Divya Sharma	91	Distinction	2	Karan Patel	55	Pass/Fail	3	Meera Nair	42	Pass/Fail	4	Sahil Gupta	88	Distinction	5	Anjali Singh	35	Pass/Fail	6	Rohan Verma	67	Merit	7	Pooja Yadav	92	Distinction	8	Vivek Kumar	71	Merit	9	Nisha Agarwal	58	Pass/Fail																																																							
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BONUS	Sort the DataFrame by Score in descending order. Reset the index. Show the top 5 students.	. BONUS Task: Sort the DataFrame by Score in descending order, reset the index, and show the top 5 <pre>bonus_df = (df.sort_values(by="Score", ascending=False) .reset_index(drop=True) .head(5)) print(bonus_df)</pre><table><thead><tr><th>StudentID</th><th>Name</th><th>Age</th><th>City</th><th>Department</th><th>Score</th><th>Grade</th><th>Passed</th></tr></thead><tbody><tr><td>0</td><td>5008</td><td>Pooja Yadav</td><td>23</td><td>Chennai</td><td>CSE</td><td>92</td><td>A</td><td>True</td></tr><tr><td>1</td><td>5002</td><td>Divya Sharma</td><td>22</td><td>Mumbai</td><td>ECE</td><td>91</td><td>A</td><td>True</td></tr><tr><td>2</td><td>5005</td><td>Sahil Gupta</td><td>21</td><td>Delhi</td><td>ECE</td><td>88</td><td>A</td><td>True</td></tr><tr><td>3</td><td>5001</td><td>Anjun Mehta</td><td>21</td><td>Delhi</td><td>CSE</td><td>78</td><td>B</td><td>True</td></tr><tr><td>4</td><td>5009</td><td>Vivek Kumar</td><td>21</td><td>Pune</td><td>ECE</td><td>71</td><td>B</td><td>True</td></tr></tbody></table><table><thead><tr><th>Score_Category</th></tr></thead><tbody><tr><td>0</td><td>Distinction</td></tr><tr><td>1</td><td>Distinction</td></tr><tr><td>2</td><td>Distinction</td></tr><tr><td>3</td><td>Merit</td></tr><tr><td>4</td><td>Merit</td></tr></tbody></table>	StudentID	Name	Age	City	Department	Score	Grade	Passed	0	5008	Pooja Yadav	23	Chennai	CSE	92	A	True	1	5002	Divya Sharma	22	Mumbai	ECE	91	A	True	2	5005	Sahil Gupta	21	Delhi	ECE	88	A	True	3	5001	Anjun Mehta	21	Delhi	CSE	78	B	True	4	5009	Vivek Kumar	21	Pune	ECE	71	B	True	Score_Category	0	Distinction	1	Distinction	2	Distinction	3	Merit	4	Merit																																		
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4	5009	Vivek Kumar	21	Pune	ECE	71	B	True																																																																																												
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